

AMTI (NMTTC) - 2011
RAMANUJAN CONTEST - INTER LEVEL

PART - A

1. ABCD is a rhombus in which B is obtuse. Perpendiculars are drawn from B to the sides and 'a' is the length of each perpendicular. The distance between their feet is 'b'. Area of the rhombus is
(A) $\frac{a^4}{b\sqrt{4a^2-b^2}}$ (B) $\frac{a^4}{2b\sqrt{4a^2-b^2}}$ (C) $\frac{2a^4}{b\sqrt{4a^2-b^2}}$ (D) $\frac{2a^4}{b\sqrt{4b^2-a^2}}$
2. Let n be a natural number. The number of n's for which $(n^4 + 2n^3 + 2n^2 + 2n + 1)$ is a perfect square is
(A) 1 (B) 2 (C) 7 (D) 0
3. $[x]$ denotes the greatest integer not exceeding x. If $a = 2 + \sqrt{3}$ then the value of $a^n + a^{-n} - [a^n]$ for any positive integer n is
(A) 2 (B) $\sqrt{3}$ (C) 1 (D) 0
4. The quadrilateral ABCD is inscribed in a circle. The diagonals AC and BD cut at Q. The sides DA and CB are produced to meet at P. If $CD = CP = DQ$ then the measure of $\angle CAD$ is
(A) 45° (B) 70° (C) 60° (D) 55°
5. The expression $(4n^3 + 6n^2 + 4n + 1)$ is
(A) Composite for all natural numbers n.
(B) Prime for exactly two natural numbers n.
(C) Prime for infinitely many natural numbers n.
(D) Composite for odd n and prime for even n.
6. $a_1, a_2, \dots$ is a sequence for which $a_1 = 2, a_2 = 3$ and $a_n = \frac{a_{n-1}}{a_{n-2}}$ for every natural number $n \geq 3$. The value of a_{2011} is
(A) 3 (B) $\frac{3}{2}$ (C) $\frac{2}{3}$ (D) 2
7. If 49 in base a and 94 in base b represent the same number in base 10, then the least possible value of this number is (in base 10)
(A) a square number (B) a four digit number necessarily
(C) a three digit square free number (D) a three digit prime
8. a, b are positive integers. If $21ab^2$ and $15ab$ are perfect squares, the minimum value of $a + b$ is
(A) 56 (B) 65 (C) 23 (D) 42
9. ABC is a triangle. $AB = 6$ cm, $BC = 9$ cm, $CA = 7$ cm. Circles are drawn with centres at A, B, C. The circles with centres A, B touch externally. The circle with centre C touches these two circles internally. The sum of the radii of these circles (in cm) is
(A) 22 (B) 17 (C) 44 (D) 111
10. ABCD is a convex quadrilateral in which $AD = \sqrt{3}$, $\angle A = 60^\circ$, $\angle D = 120^\circ$ and $AB + CD = 2AD$. M is the midpoint of BC. Then $DM =$
(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{3}{2}$ (C) $\frac{2}{3}$ (D) $\frac{2}{\sqrt{3}}$
11. A triangle has integer sides. One side is three times a second side. The length of the third side is 15. The greatest possible perimeter of the triangle is
(A) 43 (B) 46 (C) 48 (D) 52

12. If $\frac{x}{3y} = \frac{y}{2x-5y} = \frac{6x-15y}{x}$ and the expression $(-4x^2 + 36y - 8)$ takes the maximum value for $x = m$ and $y = n$ then $m + n =$
 (A) 2011 (B) 2 (C) 3 (D) 17
13. ABC is a triangle. Perpendiculars BM and CN are drawn to the tangent at A to the circum circle of ΔABC . The tangent at A meets BC produced at D. If $BC = 5$ cm, the shorter perpendicular $CN = 6$ cm and $AD = 5\sqrt{6}$ cm, then the area of the trapezoid BCNM (in cm^2) is
 (A) 30 (B) 40 (C) 35 (D) 45
14. The number of integers λ for which the equation $x^3 - 3x + \lambda = 0$ has integer roots is
 (A) 1 (B) 2 (C) 7 (D) infinite
15. A straight line passing through $A(-1, -25)$ and parallel to the line $5x - 4y + 95 = 0$ cuts the x and y axes at B and C respectively. Consider the unit grid in the plane. The number of squares in the grid containing the line segment BC in their interior is
 (A) 0 (B) 15 (C) 27 (D) 5

PART - B

1. The function $f(x)$ satisfies the condition $(x - 2)f(x) + 2f\left(\frac{1}{x}\right) = 2$ for all $x \neq 0$. Then the value of $f(2)$ is _____.
2. Let $s(x) = 6x^5 + 5x^4 + 4x^3 + 3x^2 + 2x + 1$. The zeros of $f(x)$ are a, b, c, d, e . Then the value of $(1 + a)(1 + b)(1 + c)(1 + d)(1 + e) =$ _____.
3. A quadrilateral inscribed in the circle has side lengths $\sqrt{20}, \sqrt{99}, \sqrt{22}$ and $\sqrt{97}$ in order. Taking $\pi = \frac{22}{7}$, the area of the circle is _____.
4. The number of rectangles that can be obtained by joining four of the 14 vertices of a 14 sided regular polygon is _____.
5. Let $a_0 = 1, a_{n+1} = 5a_n + 1$ for $n \geq 1$. The remainder when a_{2011} is divided by 13 is _____.
6. ABCD is a rhombus of side 2 units. $\angle B = 30^\circ$. Then the area of the region within the rhombus such that every point in this region is closer to vertex B than to vertices A, C and D is _____.
7. In a rectangle ABCD where $AB = 6, BC = 3$, point P is chosen on AB such that $\angle APD = 2\angle CPB$. Then $AP =$ _____.
8. The perimeter of the octagon formed by the roots of the polynomial equation $Z^8 - 256 = 0$, plotted in the complex plane in order is _____.
9. Deleted.
10. If $f(x) = \frac{3x}{5x+4}$ and $f(g(x)) = x$, then the expression for the function $g(x)$ is _____.
11. ABC is a triangle. P is any point inside the triangle. d_1, d_2, d_3 are the lengths of the perpendiculars to the sides BC, CA, AB respectively from P. h_1, h_2, h_3 are the altitudes to these sides respectively. The numerical value of $\frac{d_1}{h_1} + \frac{d_2}{h_2} + \frac{d_3}{h_3}$ is _____.
12. The equal sides of an isosceles triangle are each equal to 2011 cm. The length of the third side which makes the area of the triangle maximum is _____.
13. Both the roots of $x^2 - 63x + k = 0$ are prime numbers. The sum of the digits of k is _____.
14. The number of ordered pairs of integers (x, y) satisfying $x + y = x^2 - xy + y^2$ is _____.
15. ABC is a triangle in which $C = 90^\circ$. D lies on the segment BC such that $BD = AC\sqrt{6}$. E lies on the segment AC such that $AE = CD\sqrt{6}$. The acute angle between the lines AD and BE is _____.
